UI/UX DESIGN & SCREEN FLOW

Patient application, caregiver application and complete interaction specification

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

Mridula J B, Navaneeth S Narayan, Neeraj Ram B, Sonal Santhosh

Prof. Binu Rajan M R, Assistant Professor, Dept. of CSE, SCTCE

1. Purpose

This document defines the user-facing structure of the MCI assistive application before implementation. It translates the requirements into screens, navigation, actions, states and interaction rules. The patient interface prioritizes simplicity, recognition, reminders and low cognitive load. The caregiver interface prioritizes visibility, configuration, alerts and patient oversight.

2. UX Principles

- Keep the number of choices visible at one time low.
- Use large touch targets, clear labels and consistent iconography.
- Prefer plain language over technical terminology.
- Use persistent access to the most important functions: Assistant, Reminders, Schedule and Help.
- Make important information visible without requiring deep navigation.
- Never make a recognition result look more certain than it actually is.
- Use confirmation for potentially consequential caregiver actions.
- Provide obvious recovery paths when AI recognition or connectivity fails.
- Maintain the same terminology throughout the patient and caregiver experiences.

3. Application Structure

Application	Primary User	Primary Navigation	Main Goal
Patient App	Patient	Home / Assistant / Schedule / Reminders / Memory / More	Daily assistance and independent interaction
Caregiver App	Caregiver	Dashboard / Patient / Schedule / Medications / Alerts / Settings	Monitoring, configuration and safety

4. Patient App — Navigation Map

Screen ID	Screen	Entry	Main Actions
P-01	Welcome / Login	App launch	Login, account recovery
P-02	Home	Login	View today, reminders, next event, assistant
P-03	AI Assistant	Home / voice shortcut	Ask question, receive answer, access sources/context
P-04	People Recognition	Camera/Assistant	Recognize familiar person
P-05	Object Finder	Home / Assistant	Ask where an object was last seen
P-06	Schedule	Home	View day/week routine
P-07	Calendar	Home / Schedule	View appointments
P-08	Reminders	Home	View pending/completed reminders
P-09	Reminder Detail	Reminder	Acknowledge/remind later
P-10	Medication	Reminders / Schedule	View medication instruction
P-11	Memory / Conversation	Assistant / People	View permitted remembered context
P-12	Well-being	Home / More	Meditation and therapeutic tones
P-13	Location / Safety	More	View safety state if exposed to patient
P-14	Settings	More	Language, accessibility, privacy and account
P-15	Help / Emergency	Home / More	Help instructions and configured emergency action

5. Patient Home Screen — P-02

Element	Behavior
Greeting	Uses patient's preferred name and time-appropriate greeting.
Current time/date	Always visible.
Next reminder	Shows the nearest pending important reminder.
Today's schedule	Shows a compact list of upcoming activities.
Assistant button	Large primary action; opens AI assistant.
Find an object	Direct shortcut to object finder.
Who is this?	Direct shortcut to supported face recognition.
Well-being	Shortcut to meditation/therapeutic tones.
Safety state	Optional compact indicator; detailed safety information remains simple.

Home should answer three questions immediately: “What do I need to do?”, “What is next?”, and “Can I ask for help?”

6. Patient Assistant — P-03

The assistant is the central conversational entry point.

Component	Specification
Input	Text and, if implemented, voice input.
Context	Current patient, recent schedule/reminders, recognized person, object observations and permitted memories.
Answer	Short, direct response first; optional supporting context below.
Uncertainty	If information is missing or recognition is uncertain, say so instead of inventing an answer.
Follow-up	Allow the patient to ask a second question without restarting the session.
Quick prompts	Examples: “Where are my keys?”, “Who is this?”, “What do I have today?”, “When is my appointment?”

7. Person Recognition Flow

1. Patient opens “Who is this?” or the camera service detects a face in a supported context.
2. System captures/receives a supported image frame.
3. Face detection locates a face.
4. Recognition compares the representation against registered familiar people.
5. If a match passes the configured threshold, display the person's name and relationship/context if available.
6. If no reliable match exists, display “I’m not sure who this is.”
7. If the recognized person has relevant permitted memory, the assistant can use that context in the conversation.

8. Object Finder Flow

8. Patient selects “Find an object” or asks the assistant.
9. System identifies the object requested.
10. Backend retrieves the latest relevant object observation.
11. UI displays the last detected location/time and confidence/context.
12. If no observation exists, show “I don't have a recent location for that item.”
13. The system must not present an old observation as a guaranteed current location.

9. Reminder Flow

State	Patient UI
Upcoming	Reminder card with time and action.
Due now	Prominent notification/alert.
Acknowledged	Marked completed/acknowledged.

Snoozed	Shows next reminder time.
Missed	Clearly marked missed; caregiver notification may be triggered according to configuration.

10. Schedule & Calendar Flow

- Default view opens to today.
- Upcoming event is visually emphasized.
- Completed routine items remain visible but visually secondary.
- Appointments show time, title and location if available.
- Medication events are visually distinguishable from general activities.

11. Well-being Screen — P-12

Feature	Behavior
Meditation	Selectable guided session with duration.
Therapeutic tones	Selectable audio/tonal content.
Session state	Play/pause/stop with clear status.
Completion	Record session only if the user chooses or if project requirements define automatic recording.
Safety	Content is supportive; it is not represented as a substitute for professional treatment.

12. Caregiver App — Navigation Map

Screen ID	Screen	Main Actions
C-01	Login	Authenticate caregiver
C-02	Dashboard	Overview of patients, reminders, alerts and recent activity
C-03	Patient Overview	View one patient's status
C-04	Daily Schedule	Create/edit routines
C-05	Calendar	Create/edit appointments
C-06	Medication	Create/edit medication and timing
C-07	Reminders	Review reminder status
C-08	People	Register familiar people
C-09	Objects	Manage tracked objects / inspect last-seen observations
C-10	Location & Geofence	Configure safe zones and view latest location
C-11	Alerts	Review and acknowledge alerts
C-12	Conversation / Memory	Review permitted memory metadata/content
C-13	Well-being	Configure or review supported well-being content
C-14	Settings & Permissions	Manage access, privacy and preferences

13. Caregiver Dashboard — C-02

Widget	Information
Patient status	Selected patient and current connection/status.
Today's schedule	Upcoming activities.
Medication	Next medication event and recent acknowledgement state.
Appointments	Nearest appointment.
Alerts	Unacknowledged safety/system alerts.
Location	Latest location if enabled and authorized.
Recent activity	Recent relevant application events.
Quick actions	Add medication, appointment, schedule item, person or geofence.

14. Caregiver Patient Overview — C-03

The caregiver should be able to move from overview → specific detail without navigating through unrelated screens.

- Patient profile and configured preferences.
- Today's schedule and next event.
- Medication status.
- Recent alerts.
- Latest location/geofence status.
- Recent object observations.
- Recent permitted conversation/memory activity.

15. Geofencing UI — C-10

Action	UI Requirement
Create zone	Map/location selection + name + radius.
Edit zone	Change name/radius/active state.
Disable zone	Require confirmation.
Exit event	Show alert with patient, zone and time.
Latest location	Show timestamp and accuracy.
No location	Clearly state that current location is unavailable rather than implying the patient is safe.

16. Common UI States

State	Required Behavior
Loading	Show simple progress indicator and retain context.
Empty	Explain what is missing and give a next action.
Offline	Show offline state and avoid claiming live results.
AI unavailable	Provide deterministic features where possible.
Recognition uncertain	Use uncertainty language and allow retry.
Permission denied	Explain which permission is needed and why.
Server error	Show retry action and non-technical message.
Unauthorized	Do not reveal another patient's data.

17. Accessibility Requirements

- Large, readable text with scalable typography.
- High contrast between text and background.
- Buttons large enough for reliable touch input.
- Do not rely on color alone to communicate state.
- Support screen readers where the chosen framework permits.
- Use both icon and text for important actions.
- Keep interaction patterns consistent across screens.

18. Screen-to-API Mapping

Screen	Primary APIs
P-02 Home	/patients/me, /schedule, /reminders, /appointments
P-03 Assistant	/assistant/sessions, /assistant/query
P-04 Recognition	/face/recognize
P-05 Object Finder	/objects/query, /objects/{id}/last-seen
P-06 Schedule	/patients/{id}/schedule
P-07 Calendar	/patients/{id}/appointments
P-08/P-09 Reminders	/patients/{id}/reminders, /reminders/{id}/acknowledge

P-12 Well-being	/wellbeing/content, /wellbeing/sessions
C-02 Dashboard	patient/schedule/medication/reminder/alert/location APIs
C-06 Medication	/patients/{id}/medications, medication schedules
C-08 People	/patients/{id}/people, face embeddings
C-10 Location	/geofences, /locations, /alerts

19. MVP vs Later Features

MVP	Later / Extension
Login + role separation	Advanced caregiver roles
Patient Home	Personalized adaptive UI
Schedule + calendar	External calendar integrations
Medication reminders	Advanced adherence analytics
Object last-seen	More complex multi-camera tracking
Face recognition	Improved identity/context reasoning
Basic RAG memory	Long-term memory policies and summarization
Basic assistant	More advanced multimodal dialogue
Basic geofence	More sophisticated route/safety analysis
Meditation/tones	Personalized well-being recommendations

20. UI/UX Sign-off Checklist

- Every requirement has a user-facing destination or is explicitly backend-only.
- Patient and caregiver navigation are separated.
- Every major screen has loading, empty, error and offline states.
- Every action is mapped to an API endpoint.
- Privacy-sensitive screens have authorization rules.
- Recognition uncertainty is represented explicitly.
- Object locations display observation time/context.
- Reminder and medication flows are deterministic and auditable.
- Final wireframes can be produced directly from this screen list.